NETWORK PROJECT FOR NETCORE SYSTEMS LTD. USING CISCO PACKET TRACER

BY

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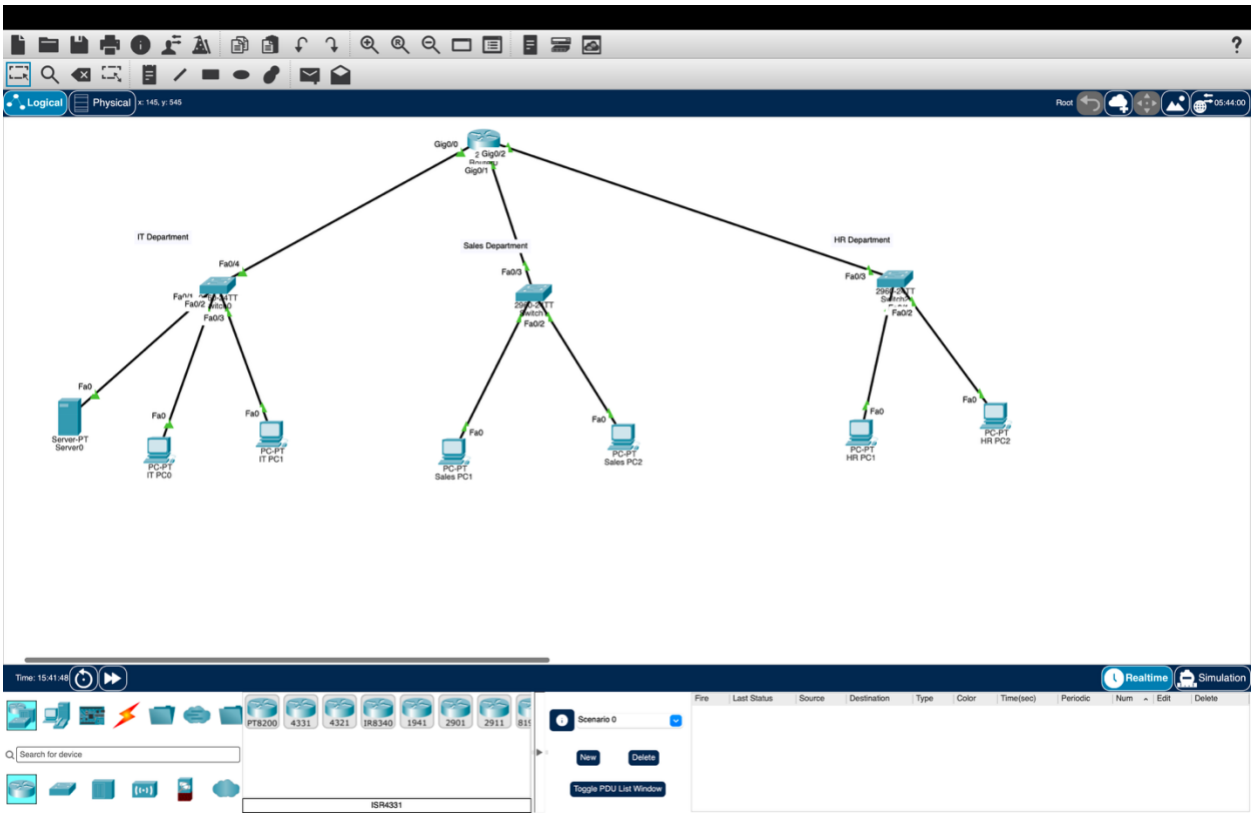


Fig 1.0 TechNova network topology

The screenshot above shows the completed network topology, with all connections functioning correctly.

To build this network, I configured one Cisco 2911 router, three Cisco 2960 switches, six PCs—two for each department—and one server. The server was installed within the IT department.

I then assigned IP addresses to all connected devices through DHCP. Each router interface was configured with the appropriate subnet address, and the correct default gateway was set on every device.

Department	IP Addresses	Devices
IT	192.168.10.1 (gateway); 192.168.10.5–10.10 (PCs); 192.168.10.2 (Server)	2 PCs & 1 server & 1 Switch
Sales	192.168.20.1 (gateway); 192.168.20.5–20.10 (PCs)	2 PCs & 1 Switch
H.R	192.168.30.1 (gateway); 192.168.30.5–30.10 (PCs)	2 PCs & 1 Switch

Fig 2.0 IP address and network overview

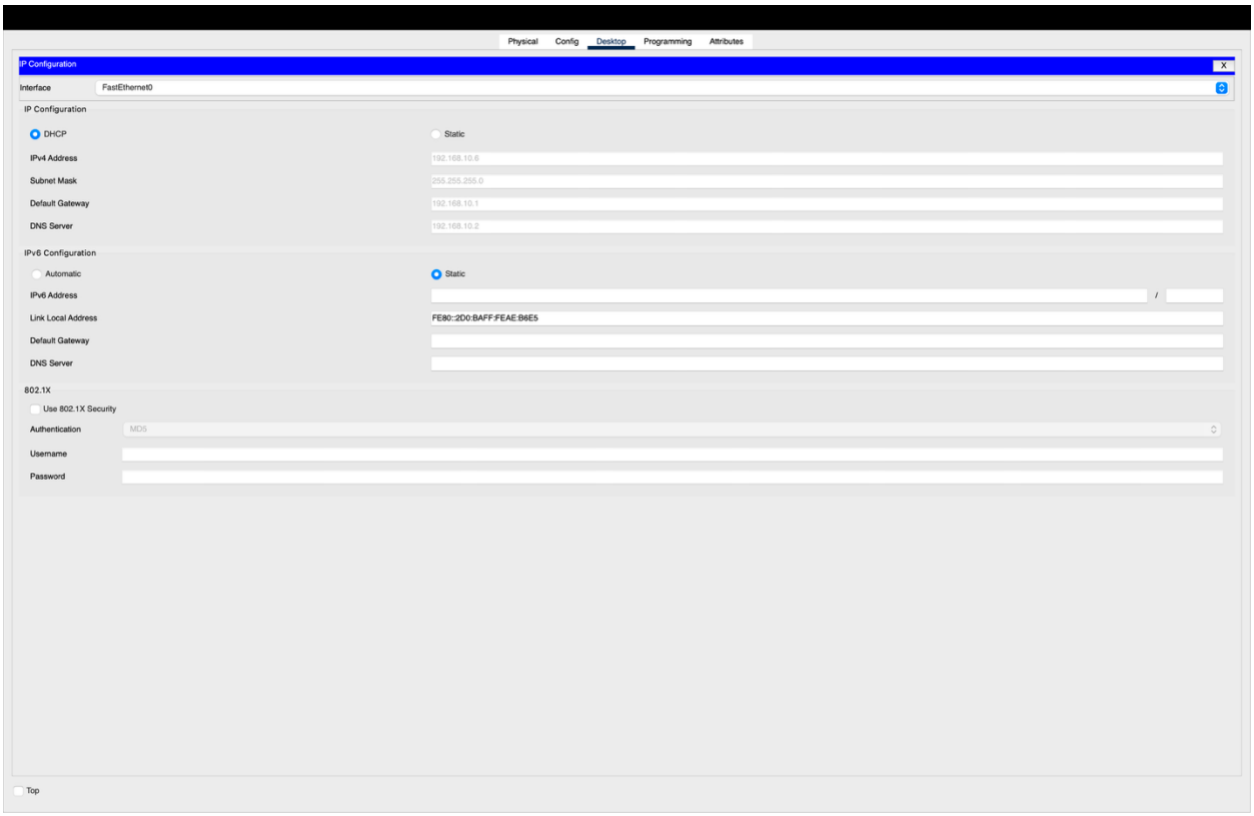


Fig 2.1 IP config for I.T PC1

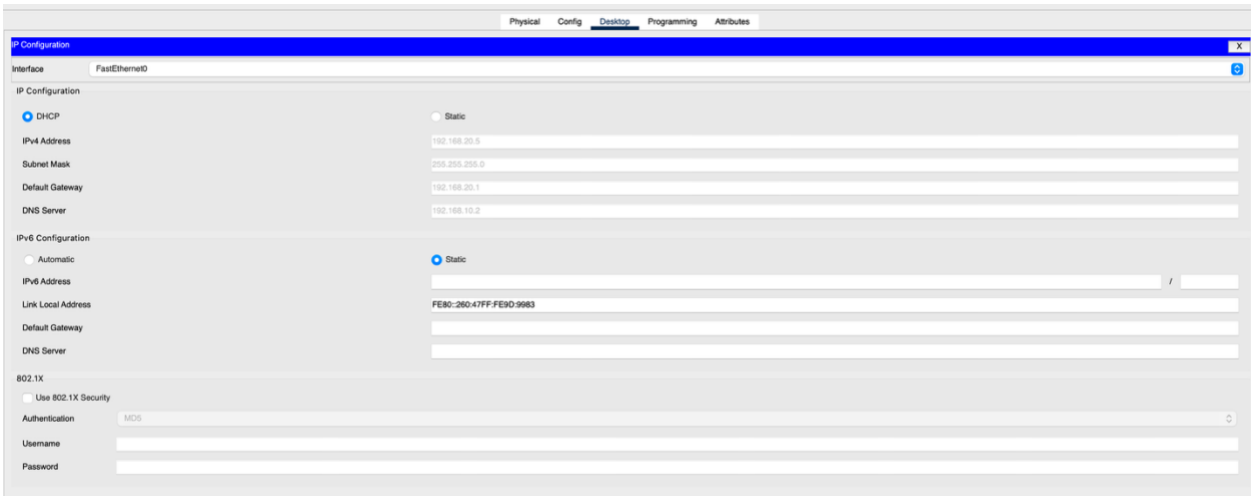


Fig 2.2 IP config for Sales PC2

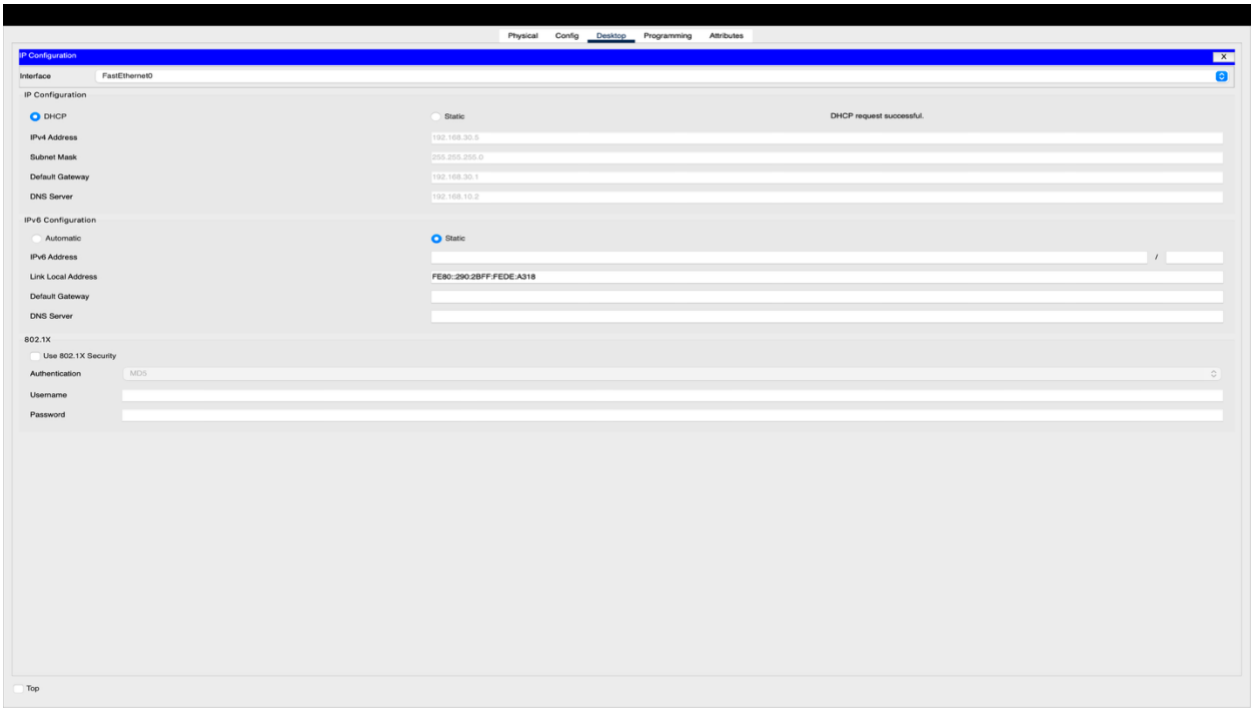


Fig 2.3 IP config for HR PC 1

I then tested basic network connectivity by using the ping command across all three departments. The results confirmed that devices in each department could communicate successfully with one another. An example of the successful connectivity test is shown below



Fig 3.0 successful HR pc ping to I.T pc

I then implemented access restrictions by configuring an Access Control List (ACL) on the router. The ACL was designed to prevent the Sales department from accessing the IT department while still allowing it to access the server's services. All other network traffic remained permitted.



Fig 4.0 Failed ping from a sales pc to an I.T pc

Below is a screenshot of the ACL command as entered in the router CLI

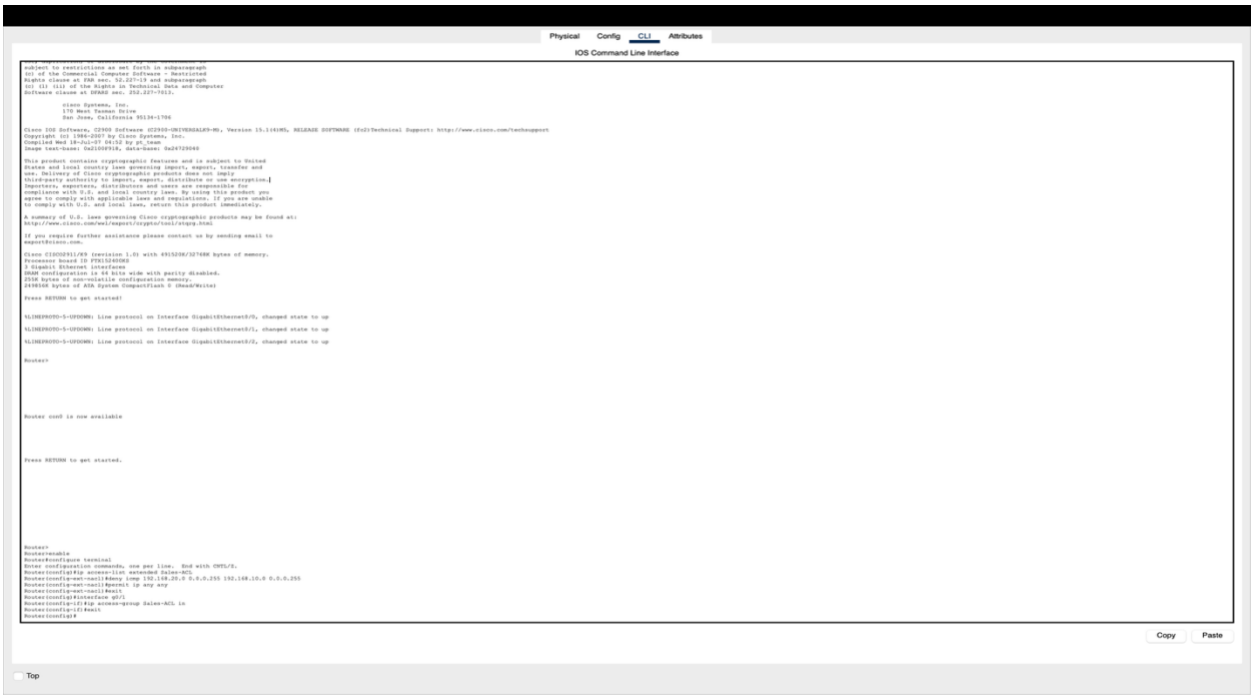


Fig 4.1 ACL command

The screenshot below confirms that the Sales department can still access the server services and communicate through all other permitted network traffic.



Fig 4.2 Sales dept. access to server